Mechanical Properties

- Are a set of physical properties which deals with application of forces to a body and reaction of the body to these forces.

Force

- It is any action which tend to produce a change in the state of rest or motion of a body i.e. force may cause one of the following reactions or all of them :

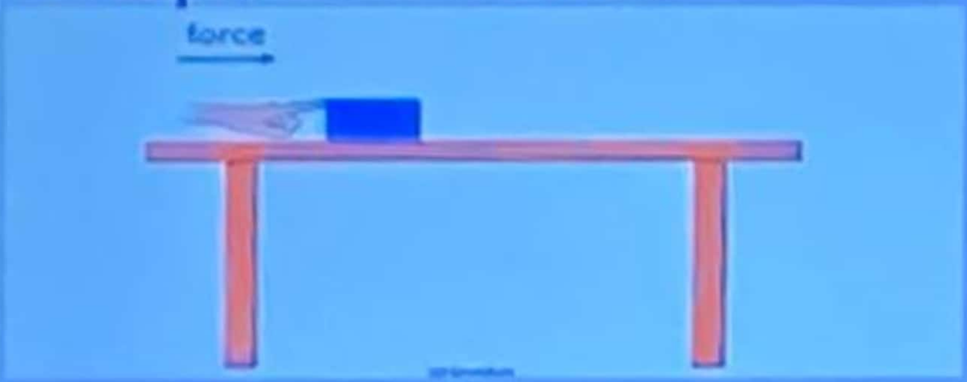
(1) Displacement.

(2) Acceleration.

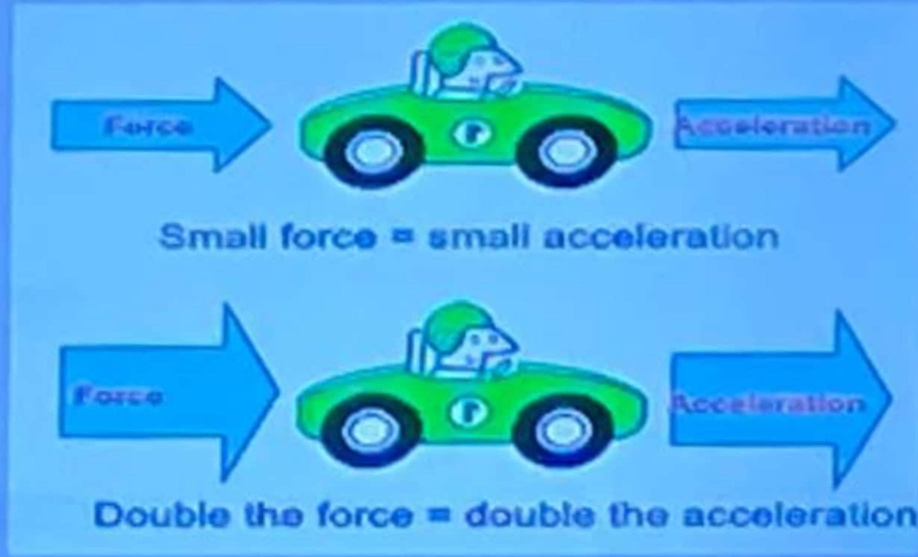
(3) Deformation

Force

Displacement.



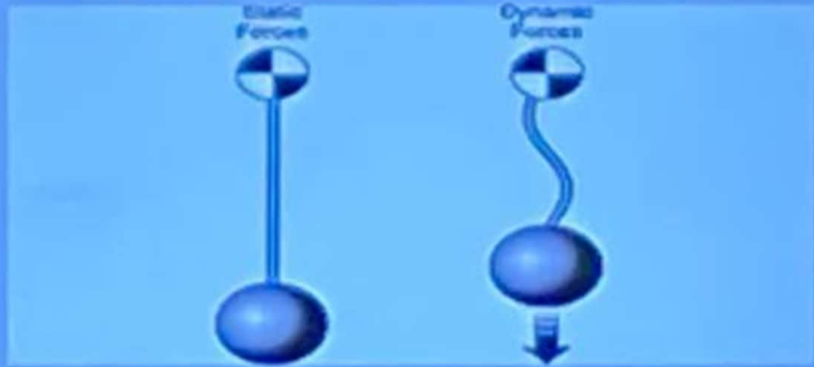
Acceleration.

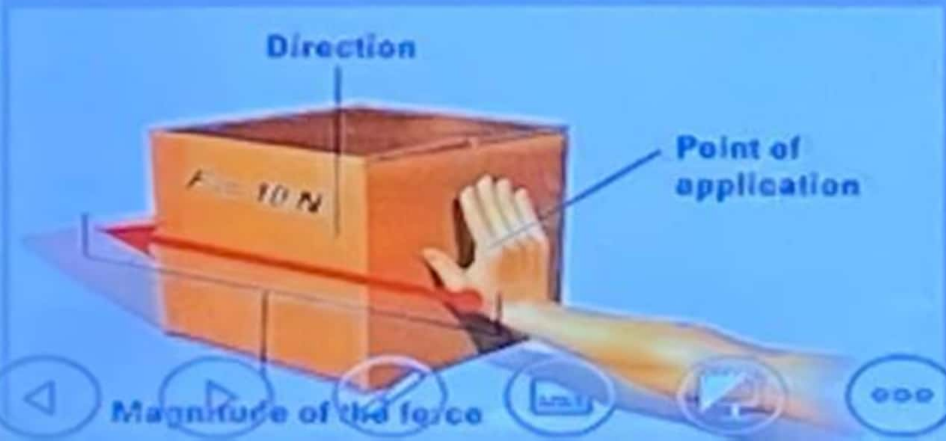


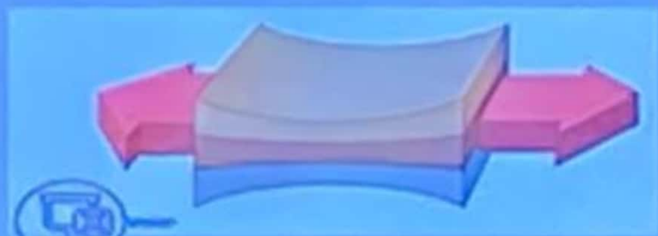
Deformation

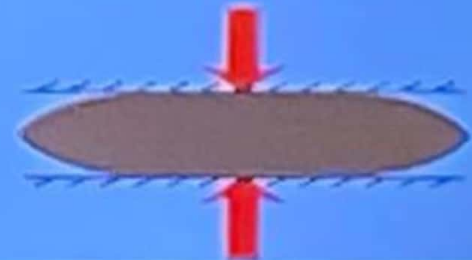
Force

- A force is defined by the following characteristics:
  a. Speed,
  b. Magnitude,
  c. Point of application
  d. Direction.



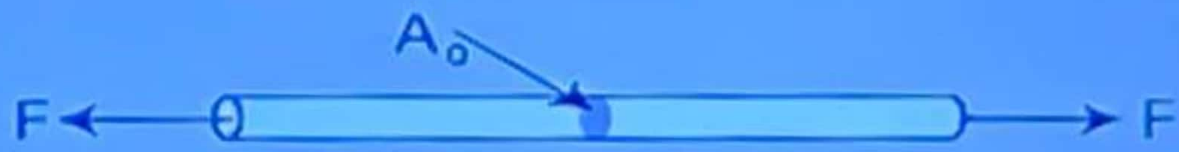






Stress

- It is the internal reaction to external force which is equal in intensity and opposite in direction to the applied force. (σ)



$$\text{Stress, } \sigma = \frac{\text{Force}}{\text{Cross-Sectional Area}} = \frac{F}{A_o}$$

- Units: $Pa = N/m^2$

  $MPa = MN/m^2 = N/mm^2$

  Kg/cm^2

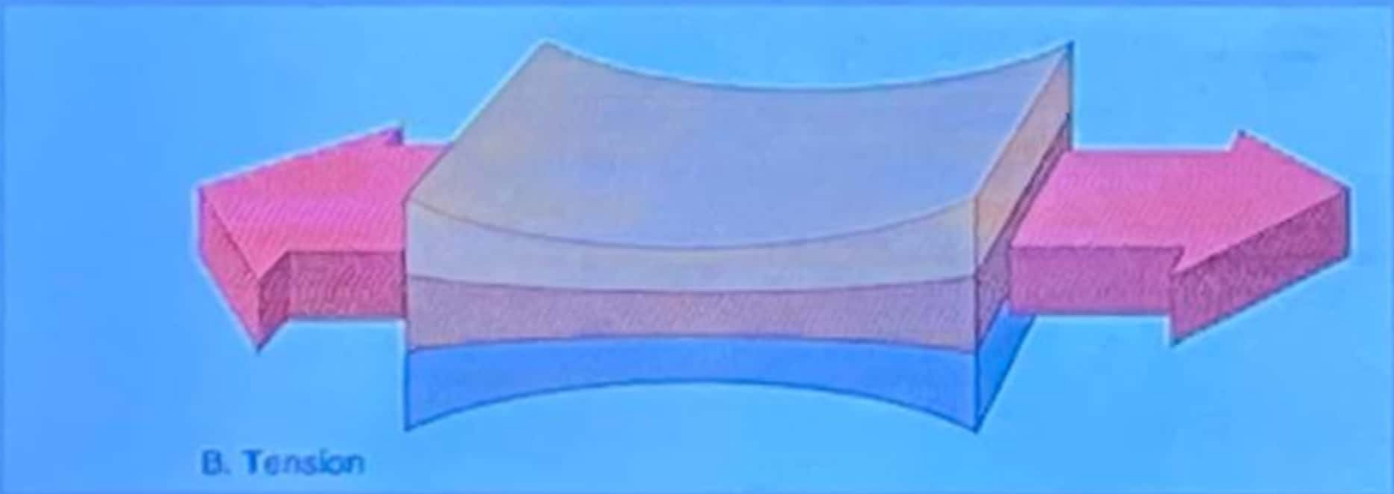
  lb/in^2

Types of stresses

- **Normal or Axial**
  - Tensile stress (σ_t)
  - Compressive stress (σ_c)
- **Tangential**
  - Shear stress (τ)
- **Complex stresses**

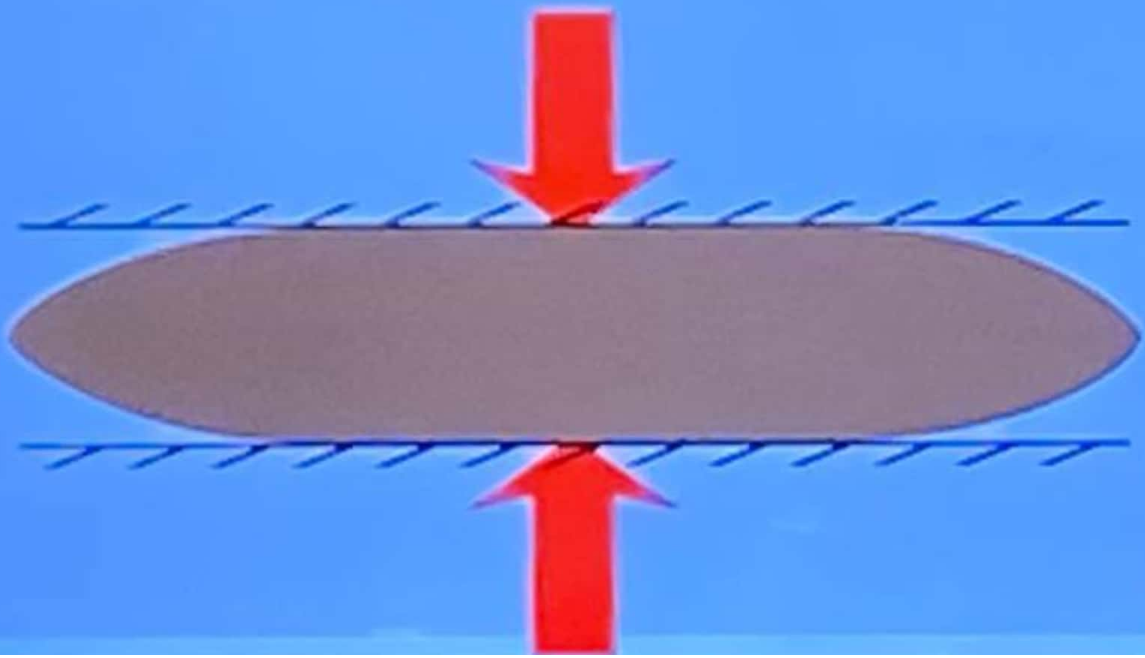
Types of stresses

- 1-Normal or Axial
- A) Tensile stress (σ_t)
  - Induced in a body loaded by tensile force i.e. two sets of axial forces directed away from each other.

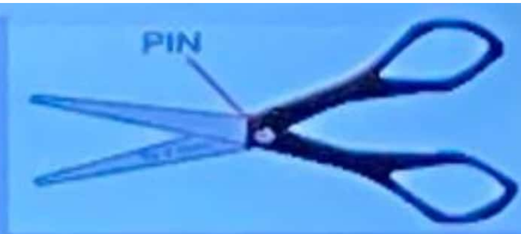


Types of stresses

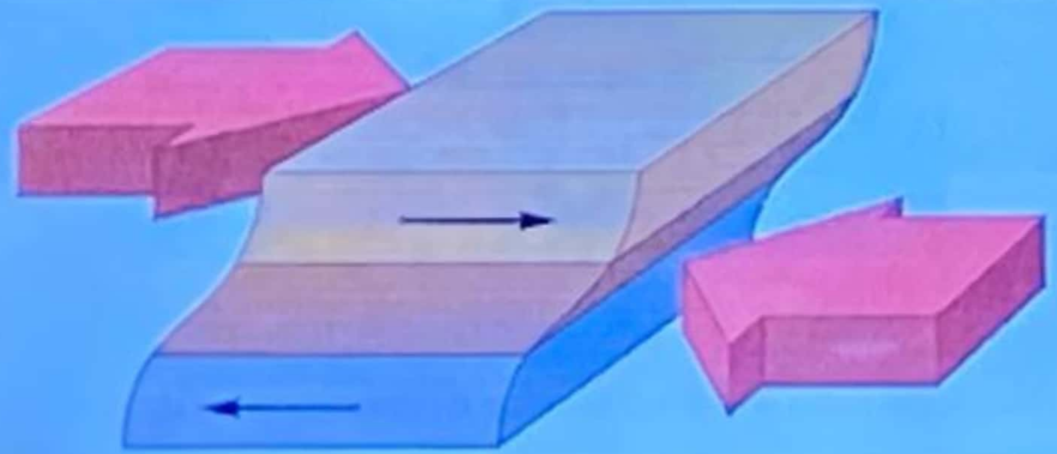
- 1-Normal or Axial
- B) Compressive stresses (σ_c)
  - Induced in a body loaded by compressive force i.e. two sets of axial forces directed towards each other.



Types of stresses



- 2) Tangential
- E.g. shear stresses (τ)
- Is the result of two sets of forces being directed towards each other or opposite each other but not in the same straight line i.e. parallel to each other (such that the distance between them will be close to zero).



Types of stresses

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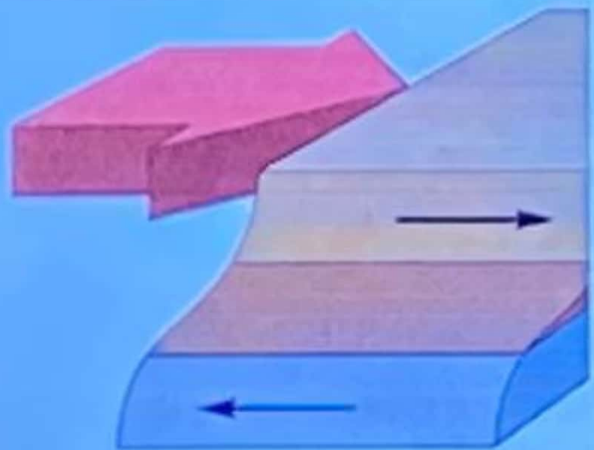
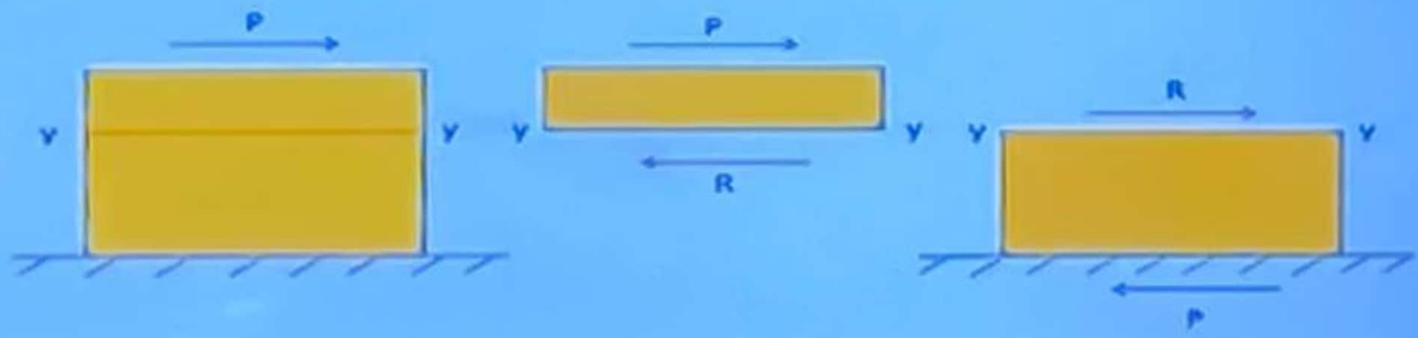
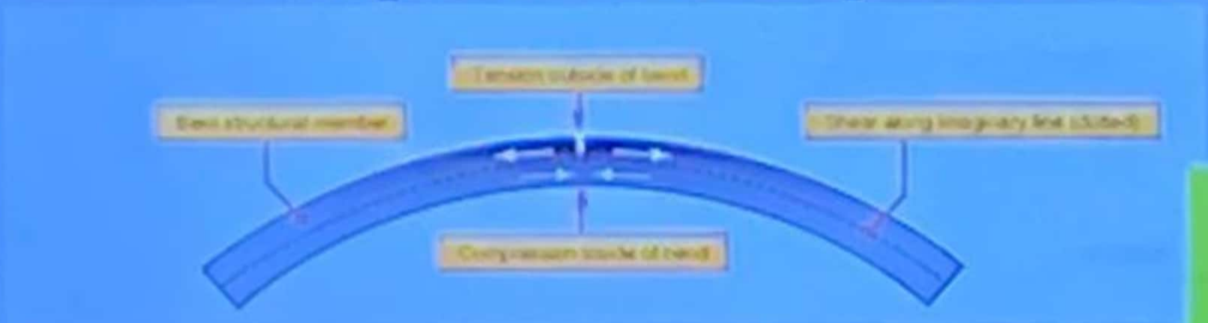
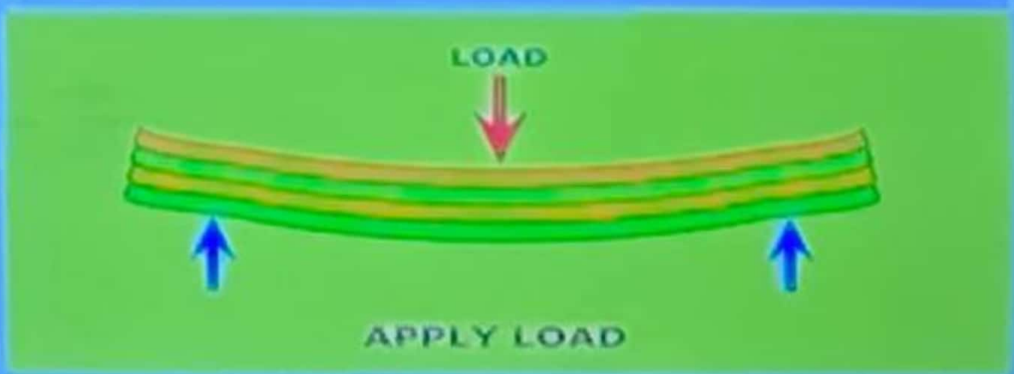




Types of stresses

- 3) Complex stresses
- If the force applied to a dental restoration is not axial, (most common practically) it may be resolved as a combination of compressive, tensile and shear stresses, or complex stresses pattern in the structure.

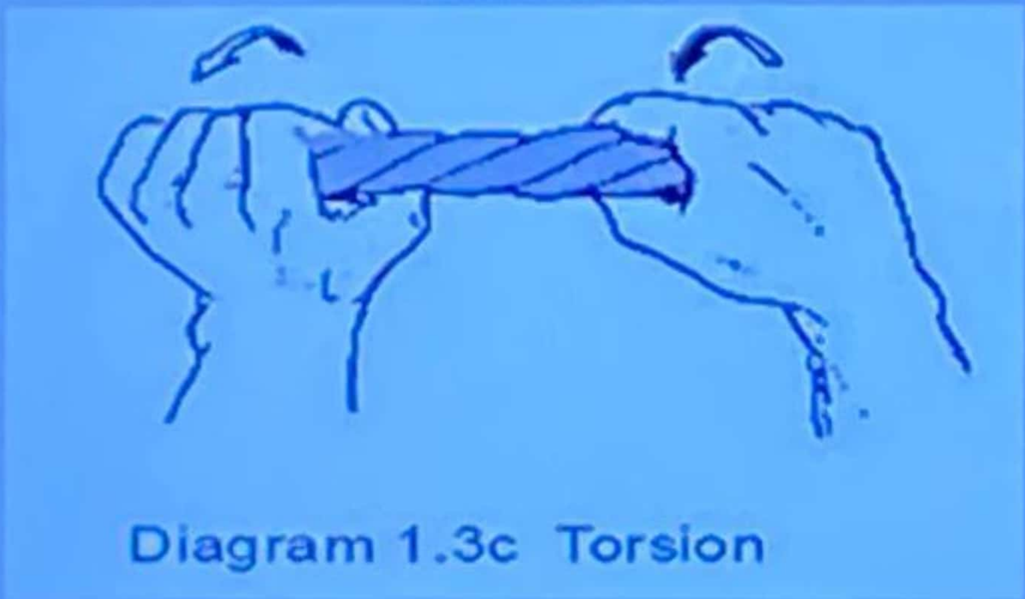




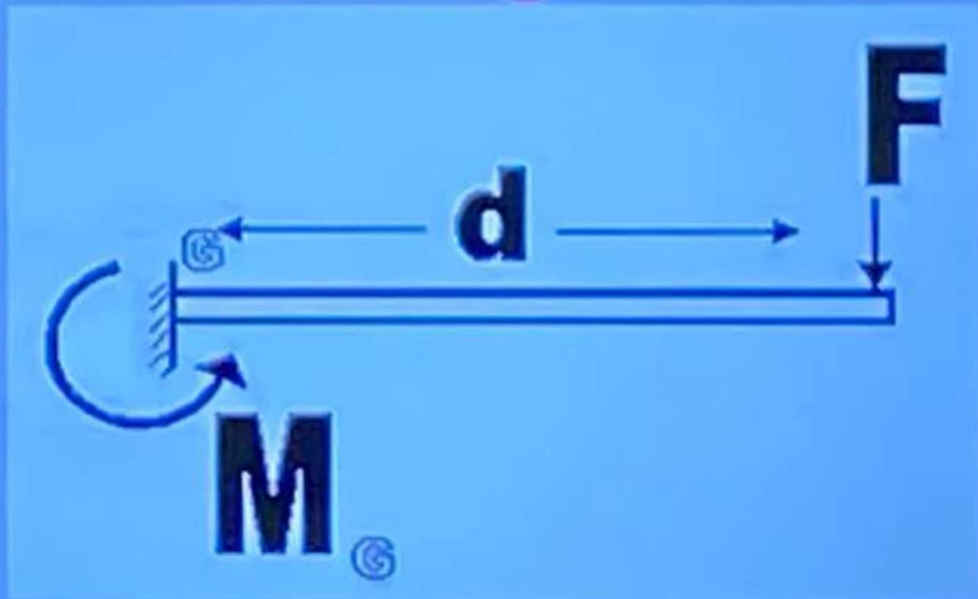
Types of stresses

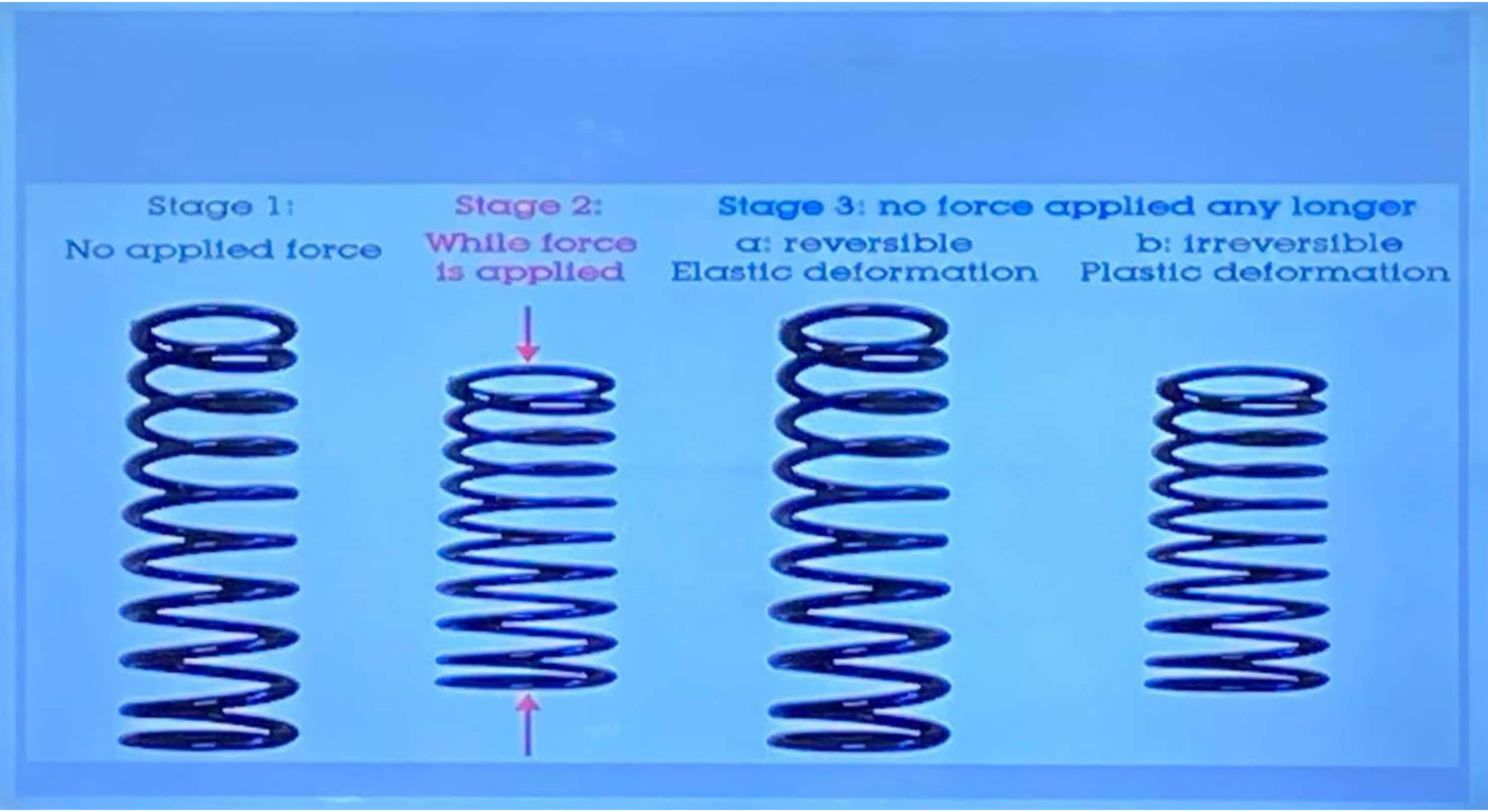
- 4) Force acting away from axis of the body.

Torsion



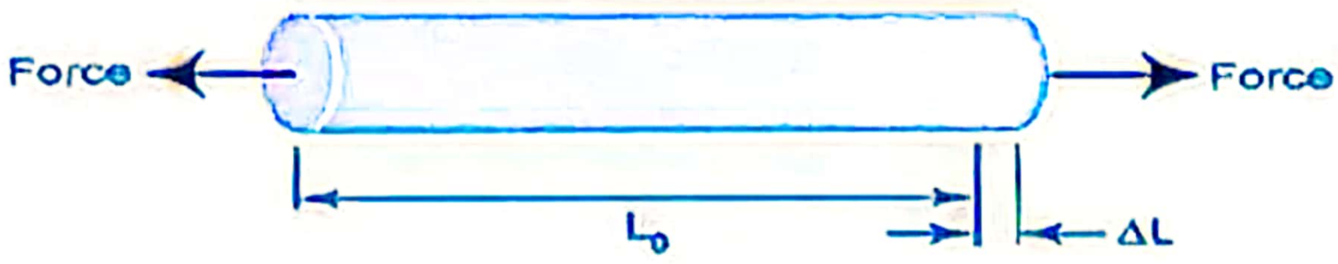
Bending




Stage 1:
No applied force
Stage 2:
While force
is applied
Stage 3: no force applied any longer
a: reversible
Elastic deformation
b: irreversible
Plastic deformation

Strain

- Strain is the change in length per unit length. (ε)



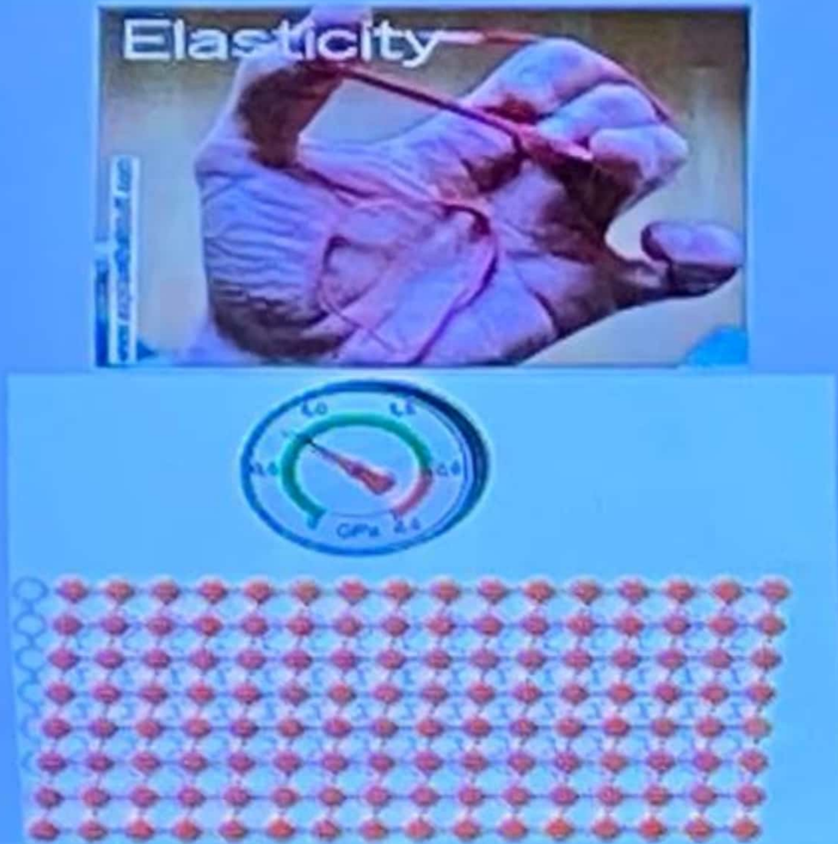
$$\text{Strain} = \frac{\text{Elongation}}{\text{Orininal Length}} = \frac{\Delta L}{L_0}$$

- Original length of 2 mm
- New length of 2.02 mm,
- It has deformed 0.02 mm
- ➢ The strain is 0.02/2 = 0.01, or 1%.

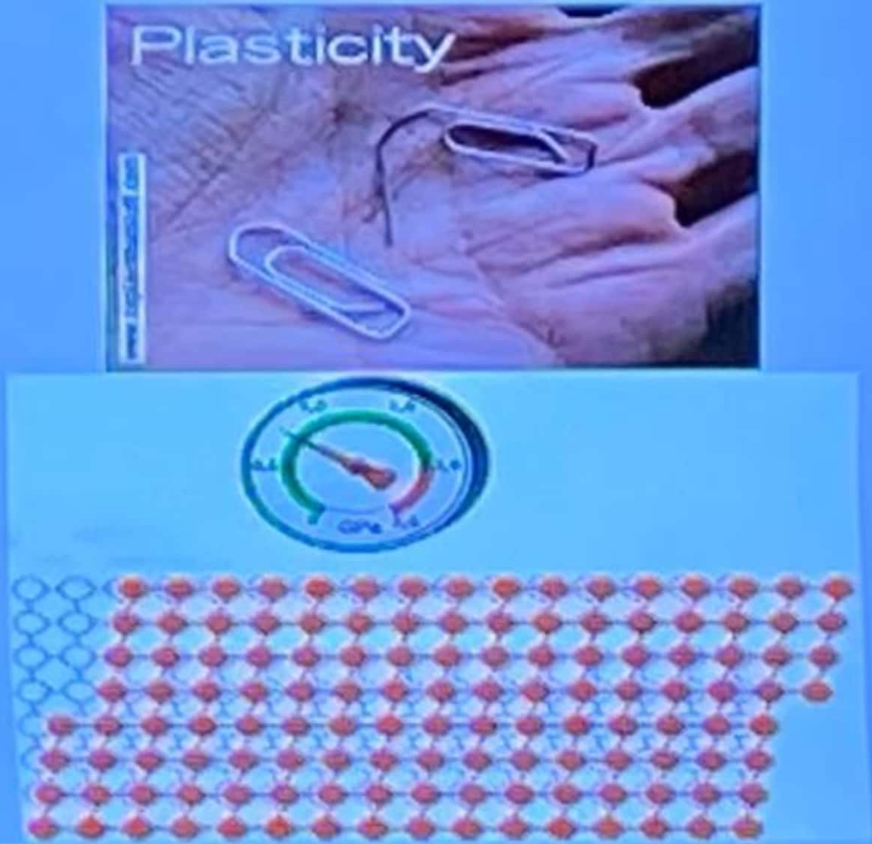
Deformation and Strain

- Any type can either be:

1) Elastic strain.

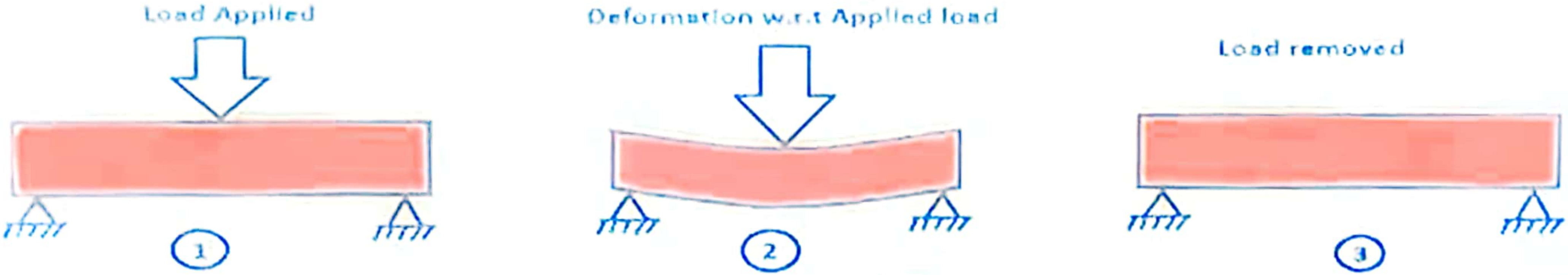


2) Plastic strain.



Strain

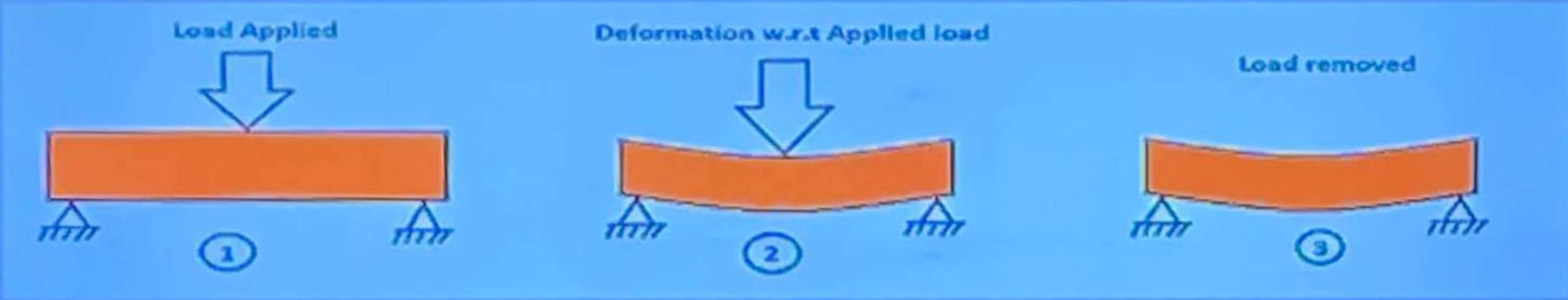
1) Elastic strain.



Elastic strain can be recovered when the force or the load is removed.

Strain

2) Plastic strain.

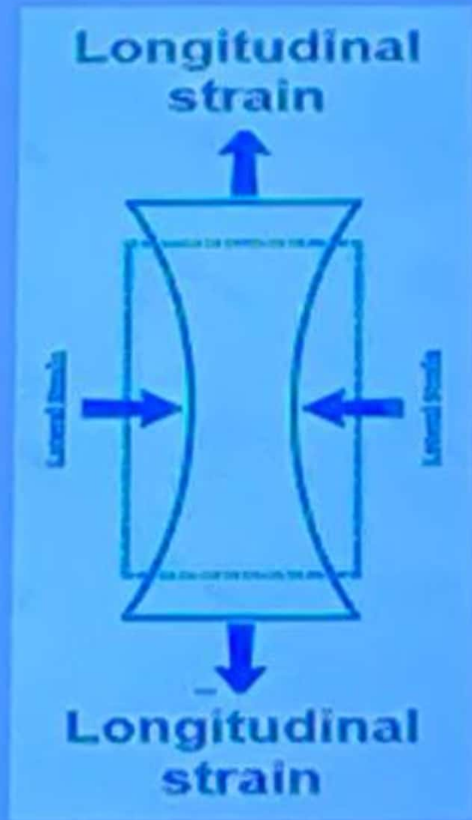


Plastic strain can not be recovered when the load is removed.

Poisson's ratio "μ"

- Within the elastic range, it is the ratio of the lateral to the axial strain.

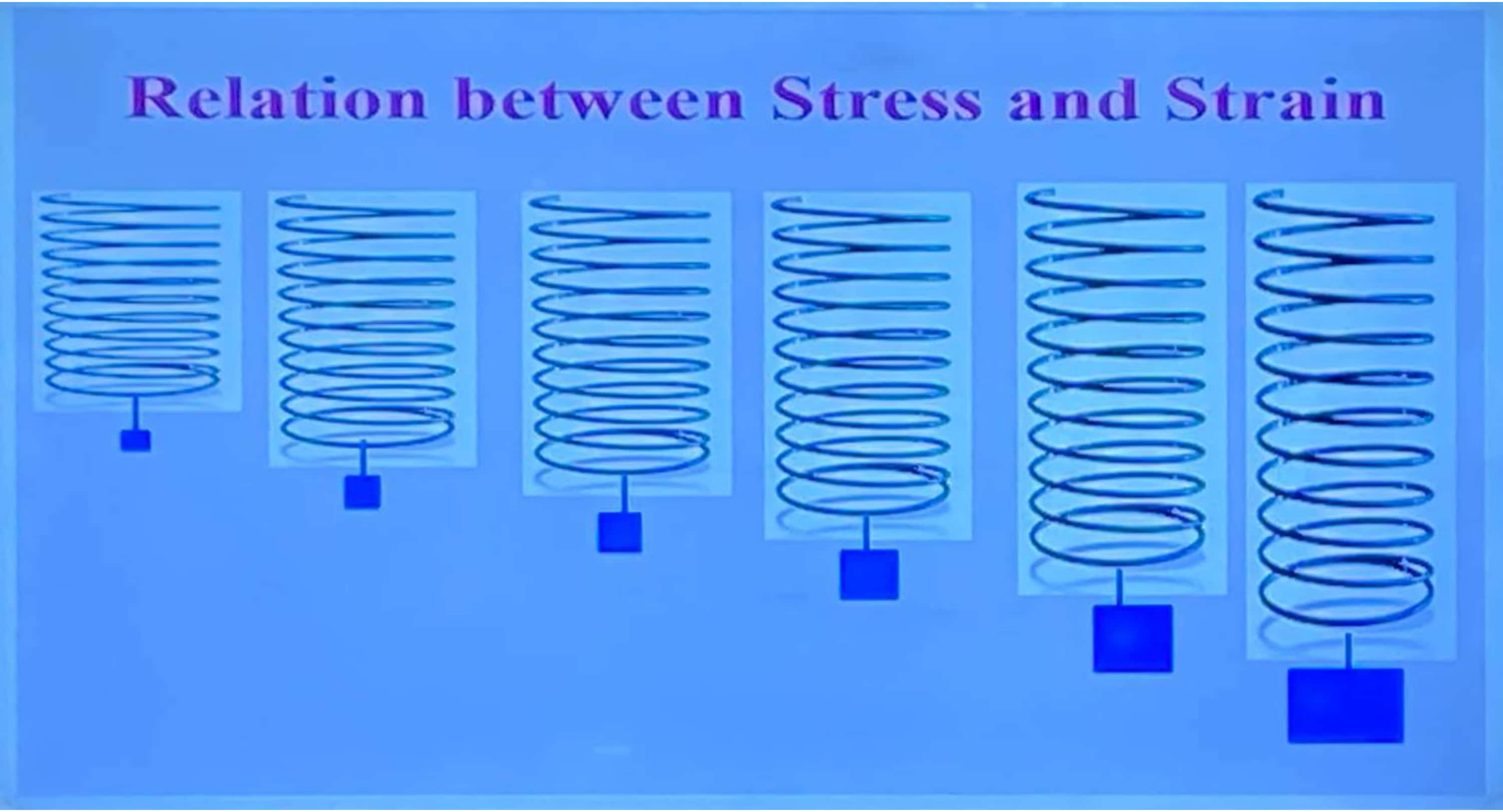
$$\text{Poisson's Ratio} = \frac{\text{Lateral Strain}}{\text{Longitudinal Strain}}$$



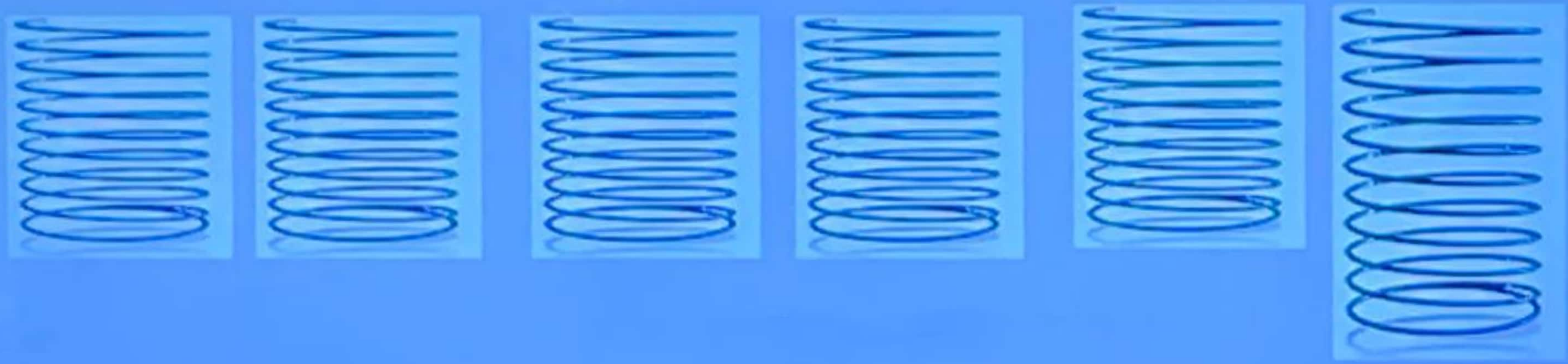
Poisson's ratio "μ"

- **Importance:**

Brittle materials shows little permanent reduction in cross section area during tensile test while ductile materials show a high degree of reduction in cross sectional area and higher Poisson's ratio.

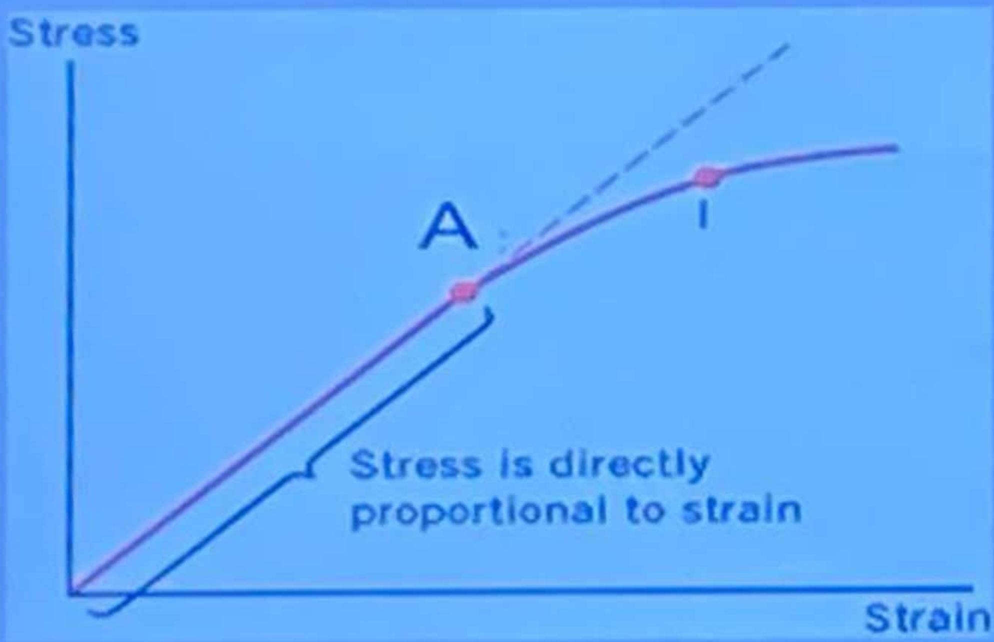

Relation between Stress and Strain

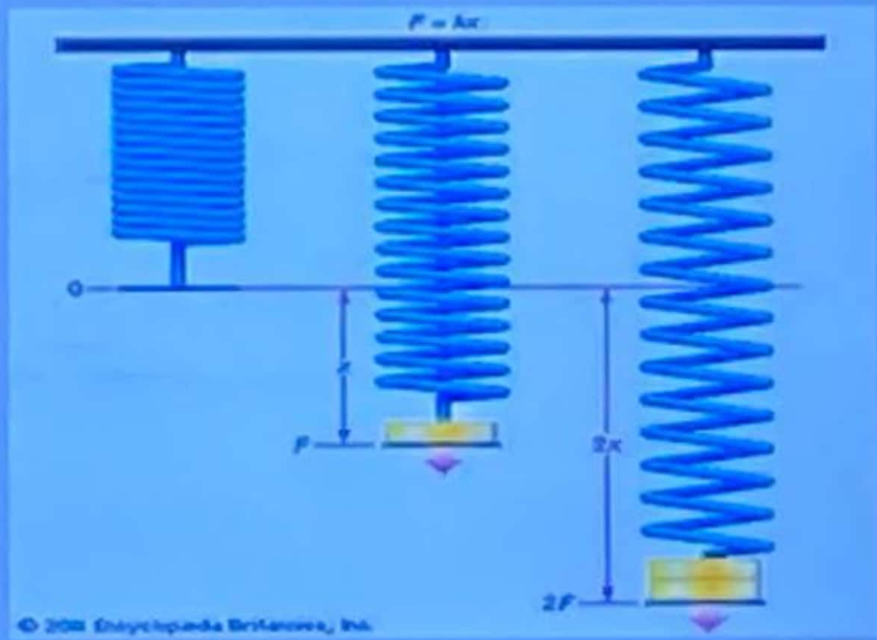
Relation between Stress and Strain



Relation between Stress and Strain

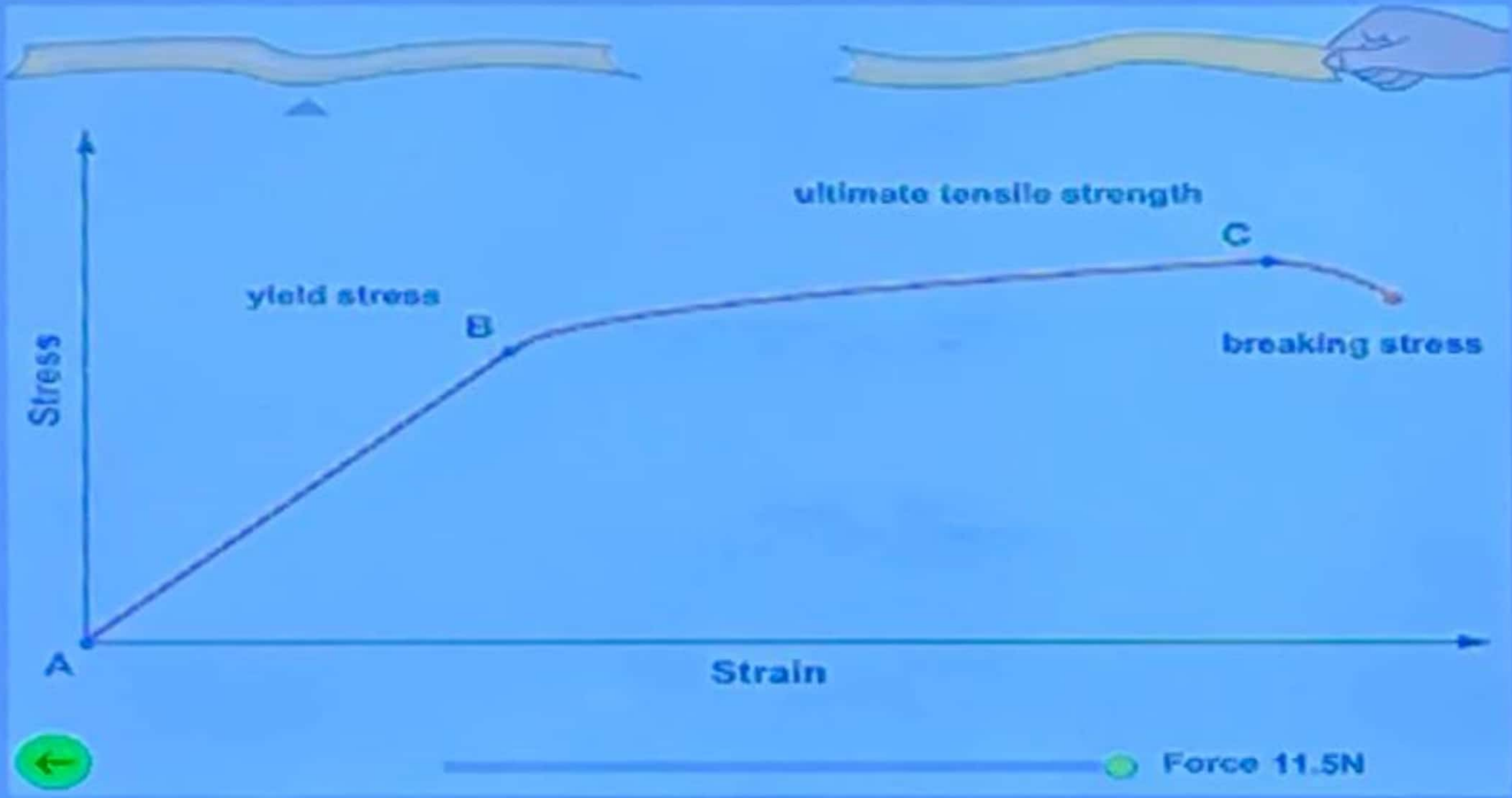
- Until a certain stress value at *A* we find that the strain is proportional to the stress induced
  - i.e. if the stress is doubled the amount of strain is also doubled



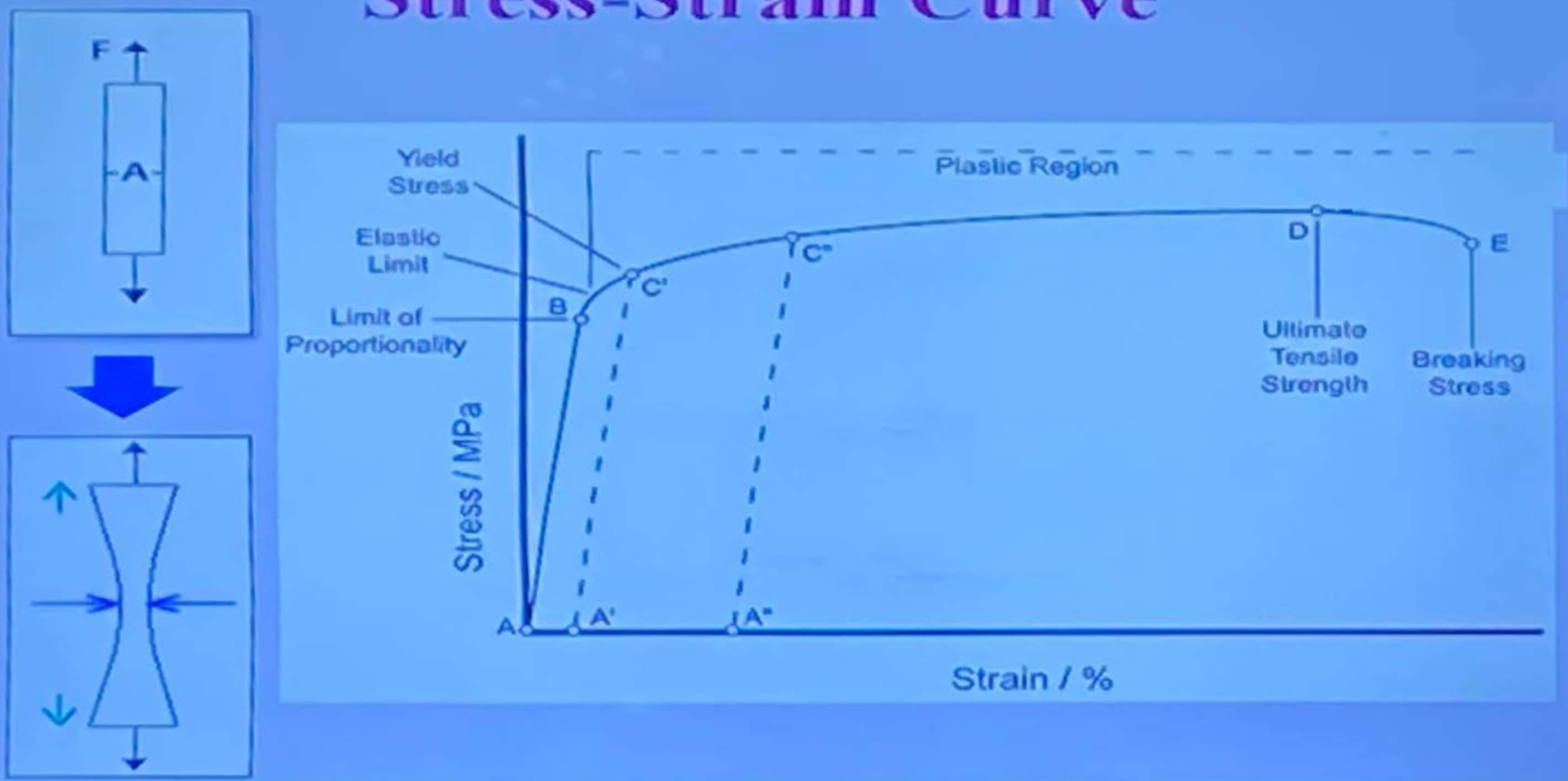


Stress-Strain Curve

- If we plot a graph for the stress and the corresponding strain of a dental material subjected to load

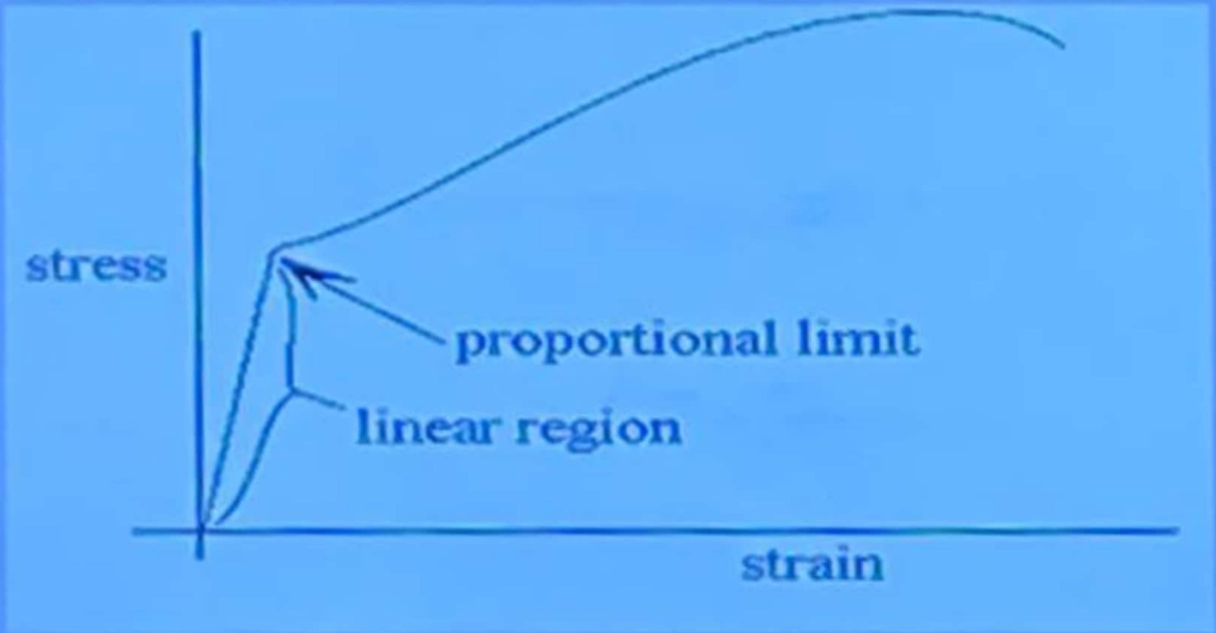


Stress-Strain Curve

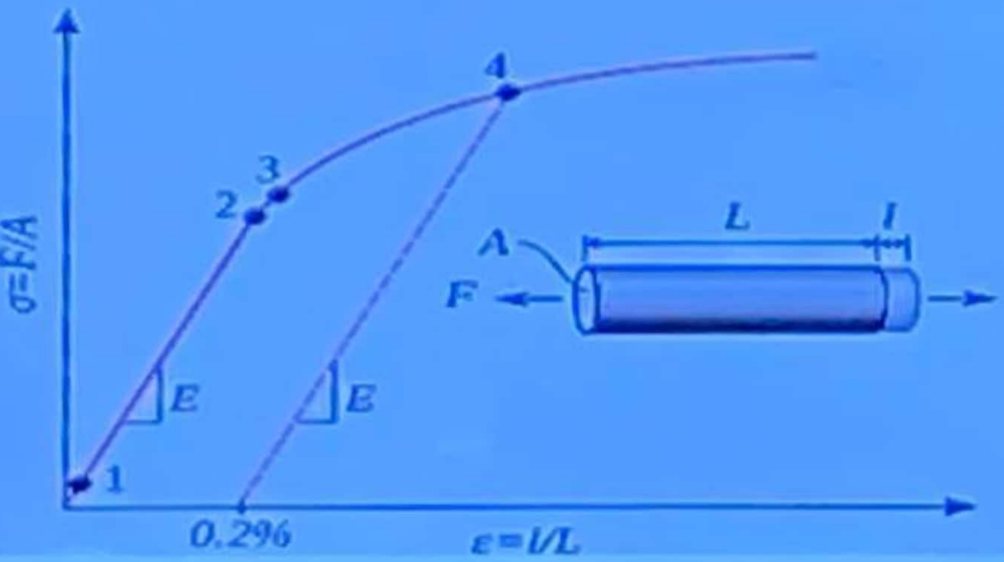


1. Proportional limit

- It is the greatest stress the material can withstand without deviation from Hooke's law or from the law of proptionality between stress and strain.



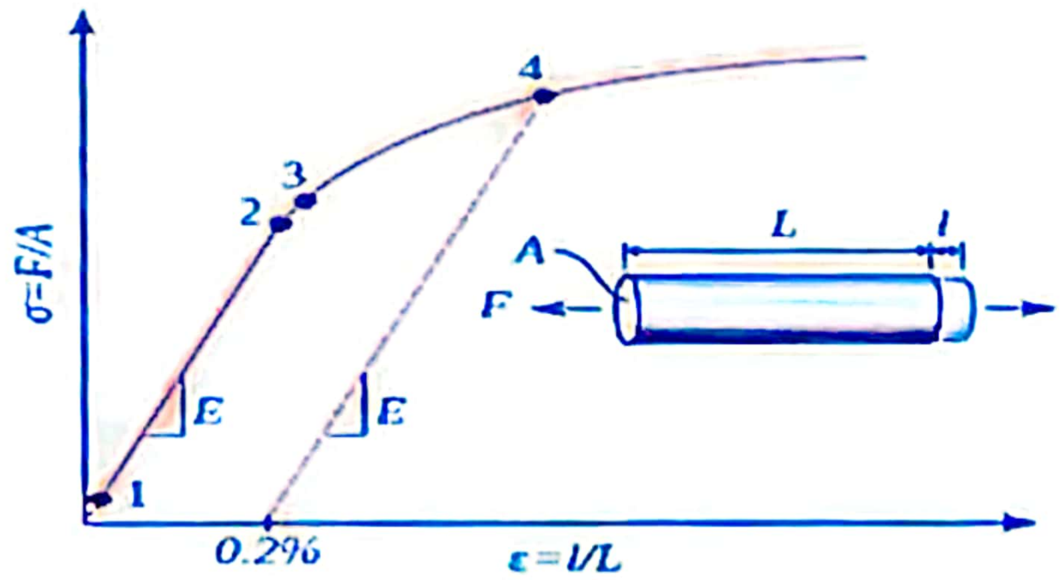
2. Elastic limit



1-2 Elastic region modulus
2. Proportional Limit.
2. Elastic Limit

3. Yield strength

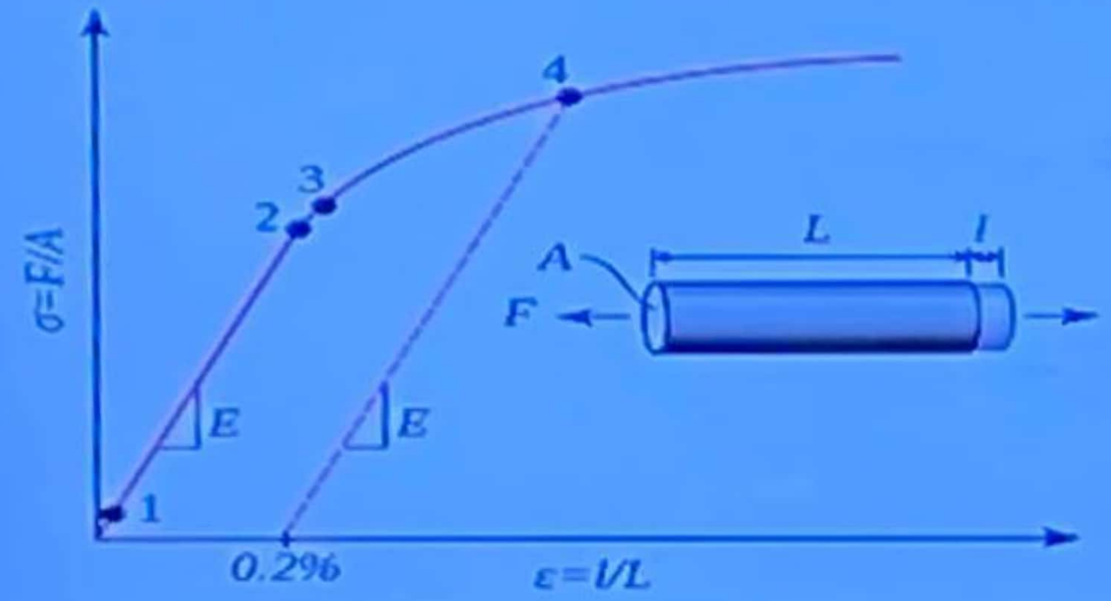
- It is the stress at which the material exhibits a specified limited deviation from law of proptionality of stress to strain (behaves in a plastic manner). (YS or σ_Y)



1-2 Elastic region modulus
2. Proportional Limit.
2. Elastic Limit
3. Yield Strength.

3. Yield strength

- The amount of permanent strain is arbitrarily selected, may be 0.1%, 0.2% or 0.5% of permanent strain, the amount of which may be referred to as ***percent offset***.



1-2 Elastic region modulus
2. Proportional Limit.
2. Elastic Limit
3. Yield Strength.
4. Yield St. at 0.2% offset.

Yield strength

Yield strength

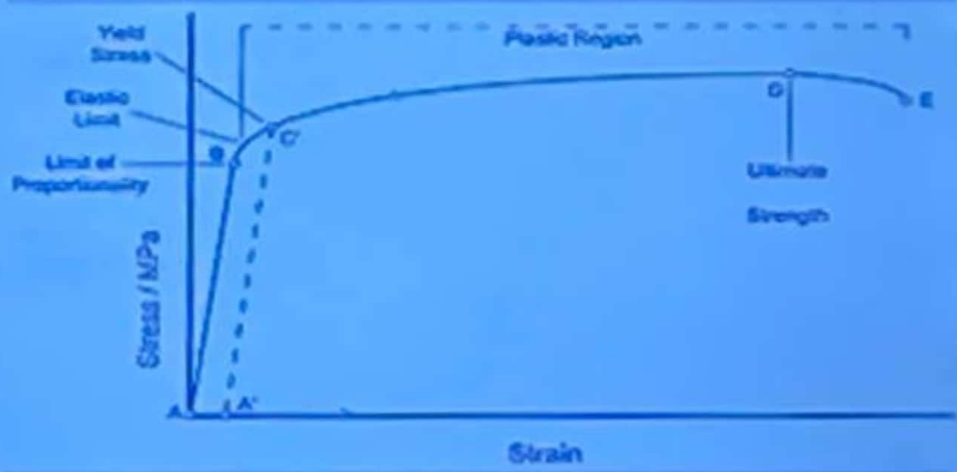
Stresses due to Masticatory Forces

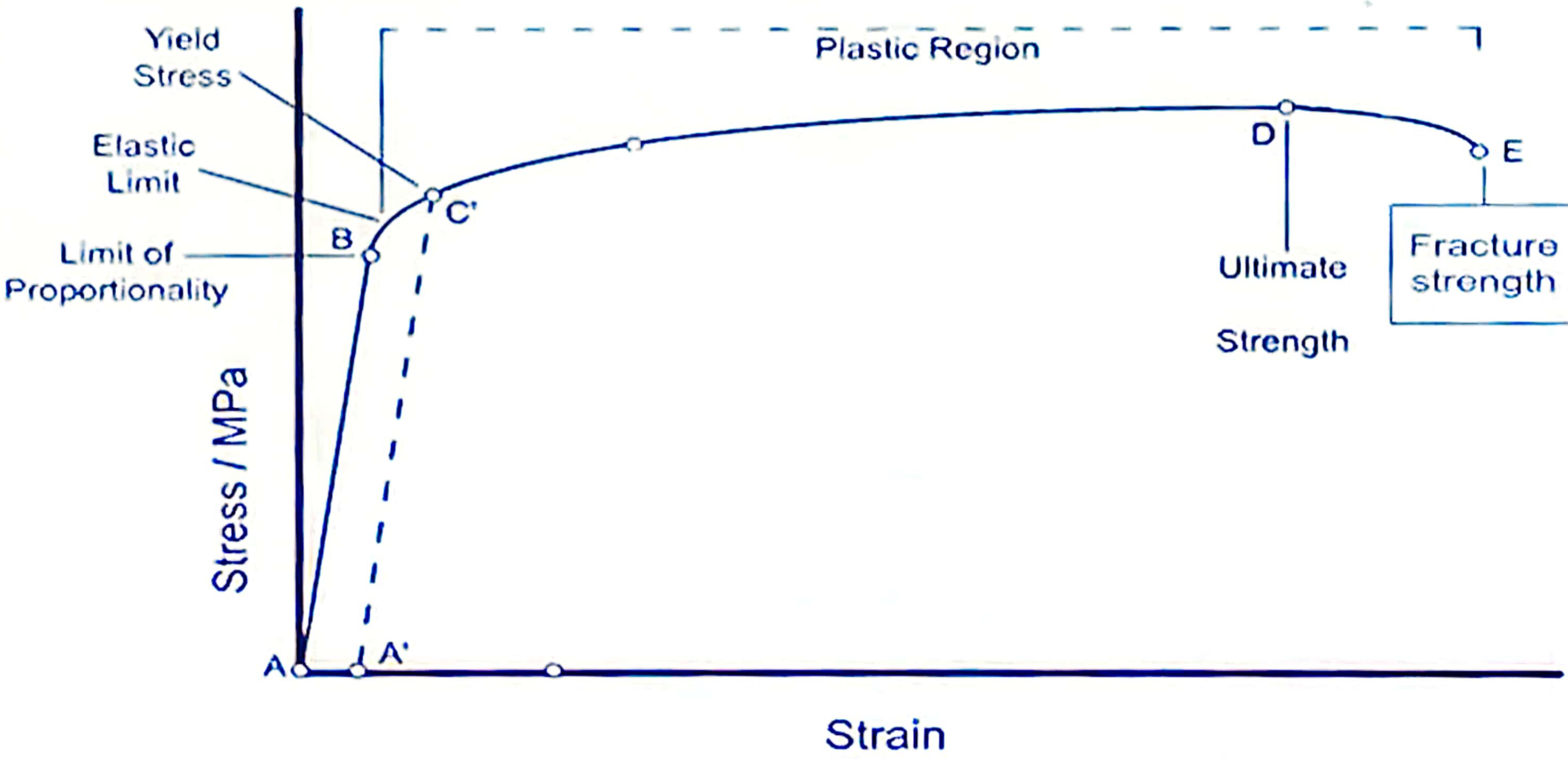
3. Yield strength

- 2- During adjustments of restorations the stress induced must be greater than their yield strength to produce permanent deformation e.g. burnishing.

4. Ultimate strength

- The maximum stress the material can withstand **before fracture**.




Yield
Stress
Elastic
Limit
Limit of
Proportionality
Plastic Region
B
C'
D
E
Ultimate
Strength
Fracture
strength
Stress / MPa
A
A'
Strain